Trudy: realistic chances for Puerto Vallarta, the path, and what it means at three addresses

The NHC's forecast and wind probabilities, the five best position forecasters of Simon (Google DeepMind's cyclone model, the ECMWF AIFS, the track consensus, the Canadian and GEFS ensemble means), three ensembles and 77 years of climatology. Least, most likely and worst cases with timing at D Terrace in Amapas, Casa Francisca in Emiliano Zapata and the house in El Pitillal. At the end, the GFS run out to 384 hours.

Tuesday 13 October 2026, 9 pm Vallarta time (14 Oct 03Z) · NHC advisory 3 · model runs of 18Z and 00Z early versions · not an official forecast

Bottom line

- **The NHC now forecasts a major hurricane at the Jalisco–Colima coast on Saturday, then north across Jalisco.** Advisory 3: Trudy is a 40-kt tropical storm at 11.2N 97.0W moving northwest at 17 km/h, forecast to be a hurricane by Wednesday night and a major hurricane by Friday, reaching 19.0N 104.4W (the coast near Manzanillo) at 6 pm Saturday at 115 kt and 21.0N 106.0W (off Nayarit, northwest of the bay) at 6 pm Sunday at 75 kt. That line passes about 38 km southwest of Puerto Vallarta around Sun 18 Oct 11:00. No watches or warnings yet; the NHC says watches will likely be needed Wednesday.
- **NHC wind probabilities for Puerto Vallarta (the number to use):** tropical-storm wind (≥ 63 km/h) **34%**, 50 kt (≥ 93 km/h) **10%**, hurricane wind (≥ 119 km/h) **2%**, all between Friday night and Sunday night. Manzanillo: 64%, 39% and 24%.
- **The five best Simon models agree with a close pass.** Their newest lines pass ECMWF AIFS 47 km, Google DeepMind cyclone model 64 km, Canadian ensemble mean 166 km, track consensus TVCN 187 km, GEFS ensemble mean 292 km, all to the southwest, Sunday morning to Monday early. Two of the five within 100 km, four of the five within 200 km, all five within 300 km; two of the five give tropical-storm wind in the city. The GFS and the ECMWF IFS are the outliers: the GFS kills Trudy over Michoacán, the IFS is two days slower.
- **Our reading:** about **4 in 5** that the center passes within 300 km (rain bands, big surf, gusty wind), about **1 in 3** within 100 km, about **1 in 3** for tropical-storm wind in the city, about **1 in 10** for 93 km/h winds, and low (NHC: 2%) for hurricane wind, because the official line crosses the Sierra before it reaches the bay.
- **When:** swell from Friday; rain bands from Saturday afternoon; the worst Saturday night to Sunday afternoon; easing Sunday night to Monday.
- **Most likely at the buildings** (NHC probabilities, median): Amapas 48 km/h gusting 67, Casa Francisca 41 gusting 60, Pitillal 37 gusting 54; 100–200 mm of rain, more on the slopes. **Worst case among the five** (Google DeepMind cyclone model 00Z): center 64 km southwest at Sun 18 Oct 08:00; Amapas 74 km/h gusting 103; about 217 mm before the mountains add theirs; Pitillal and Cuale rivers in flood.
- **What to do now:** prepare as for a storm on Sunday: water, food and medicine for three days, phones charged, documents in a bag, the car parked on high ground in Pitillal by Saturday afternoon, balcony items in at Amapas by Saturday. Watch for the Wednesday watches.

34%
NHC: tropical-storm wind (≥ 63 km/h) in Puerto Vallarta, Friday night to Sunday night

38 km
closest approach of the NHC line, Sun 18 Oct 11:00, after landfall near Manzanillo

40%
of the five best Simon models within 100 km (100% within 300 km)

Sat–Sun
rain bands from Saturday afternoon; worst Saturday night to Sunday afternoon

1. NHC advisory 3

Discussion 3 (Mahoney): Trudy has a large circulation but a compact core, with tropical-storm winds only in the northern semicircle and within 60 n mi of the center; the center was hard to place. The track "is difficult and uncertain, with higher than normal spread in the guidance from 72 h onward", as Trudy reaches weak steering between a retreating ridge and a trough over the western United States. "The deterministic global models are significantly slower than the HFIP Corrected Consensus Aid (HCCA), Google DeepMind (GDMI), the European Centre's AIFS, and the AI-GFS. The new track forecast closely follows HCCA and GDMI, which have been the best-performing track forecast aids this season." Rapid intensification is forecast. "For this cycle, the GFS track is unrealistic." Key message: "Trudy is forecast to be a major hurricane as it approaches the coast of southwestern and west-central Mexico late this week and into the weekend… watches could be required for a portion of these areas on Wednesday."

Valid (local)	Position	Wind, kt	Strength
Tue 9 pm (now)	11.2N 97.0W	40	tropical storm
Wed 6 pm	12.6N 100.0W	65	category-1 hurricane
Thu 6 pm	14.3N 101.5W	100	category 3
Fri 6 pm	16.3N 102.5W	115	category 4
Sat 6 pm	19.0N 104.4W (coast near Manzanillo)	115	category 4
Sun 6 pm	21.0N 106.0W (off Nayarit)	75	category 1

Average NHC track errors are about 100 n mi (185 km) at day 4 and 125 n mi (230 km) at day 5, and about 15 kt in intensity. Puerto Vallarta's probabilities: 34 kt 34% (14% Friday night to Saturday night, 20% Saturday night to Sunday night), 50 kt 10%, 64 kt 2%. Barra de Navidad 44 / 22 / 9%, San Blas 38 / 13 / 6%, Islas Marías 39 / 17 / 8%.

2. The five best Simon models

The five best position forecasters of Simon's last two days, leaving out the NHC's own products and the climatology baseline (12–36 h mean error in km in the second column). Their 00Z early versions, with the 18Z runs for comparison. The NHC corrected consensus, excluded here, passes 21 km SW at Sun 18 Oct 11:00.

Model	Simon km	Closest km	Side	When	kt then	Peak kt	Landfall	18Z	Amapas km/h	E. Zapata	Pitillal	Rain mm
ECMWF AIFS †	50	**47**	WSW	Sun 18 11:00	79	89	Jalisco south coast, Sun 18 09:00	93 WSW	91	79	68	145
Google DeepMind cyclone model	33	**64**	SW	Sun 18 08:00	70	110	Jalisco south coast, Sat 17 21:00	103 WSW	74	64	56	217
Canadian ensemble mean	54	**166**	SSW	Sun 18 19:00	43	52	stays offshore	199 SW	33	29	25	102
Track consensus TVCN †	51	**187**	SW	Mon 19 02:00	76	89	stays offshore	255 SSW	46	40	35	107
GEFS ensemble mean	57	**292**	SW	Sun 18 18:00	37	49	stays offshore	314 SW	17	15	13	52

Closest approach to 20.61N 105.23W, Vallarta time. Building winds: sustained, if the center followed that line exactly, from the parametric field of the Simon graphics with the Simon exposure factors and no weakening over land; not probabilities. Rain: R-CLIPER-type total at the city, no terrain. † intensity from the NHC intensity consensus. The AIFS early version is the 18Z AIFS run moved to the 00Z center by the NHC's own rule.

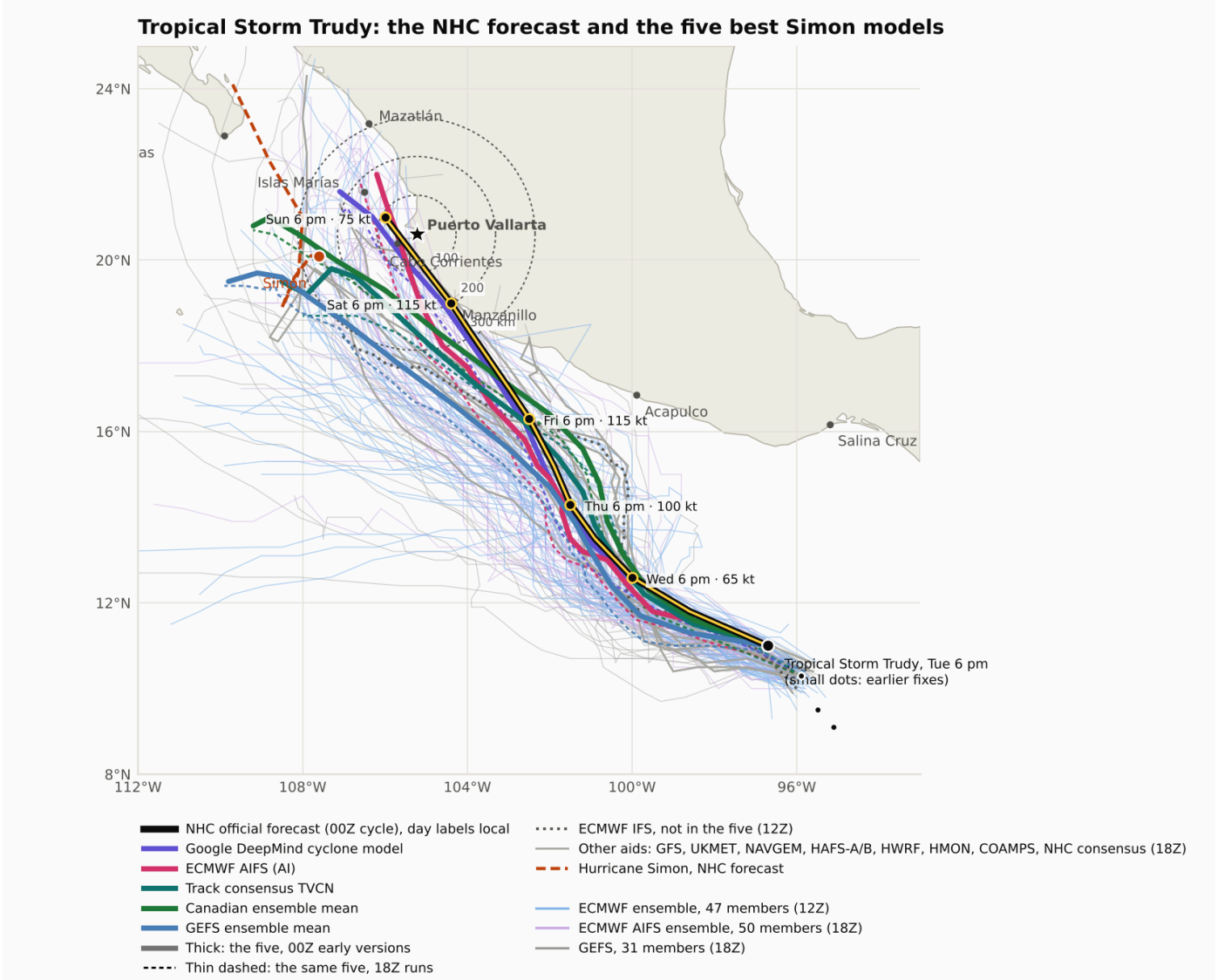


Figure 1. The NHC forecast (black and yellow, with local day and strength), the five best Simon models (thick: 00Z; thin dashed: 18Z), the other aids, the three ensembles, and Simon's NHC forecast (orange dashed).

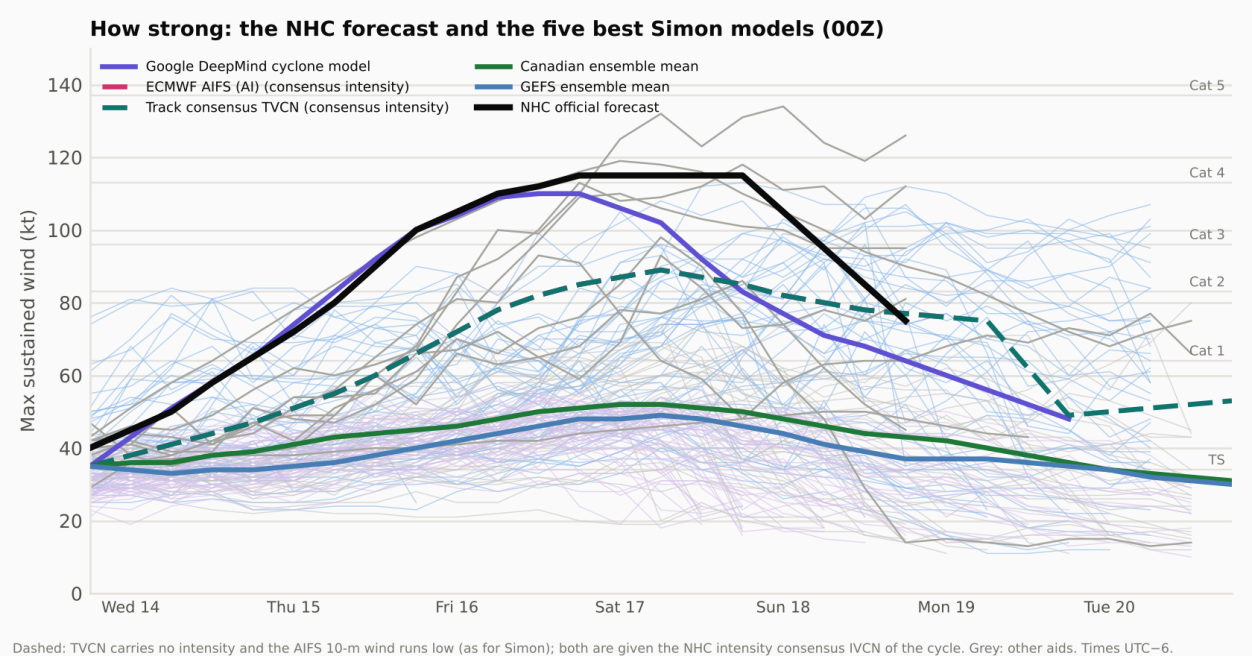


Figure 2. Intensity. The NHC forecasts a peak of 115 kt; DeepMind 110 kt; the ensemble means are smeared and weak by construction.

3. The chances

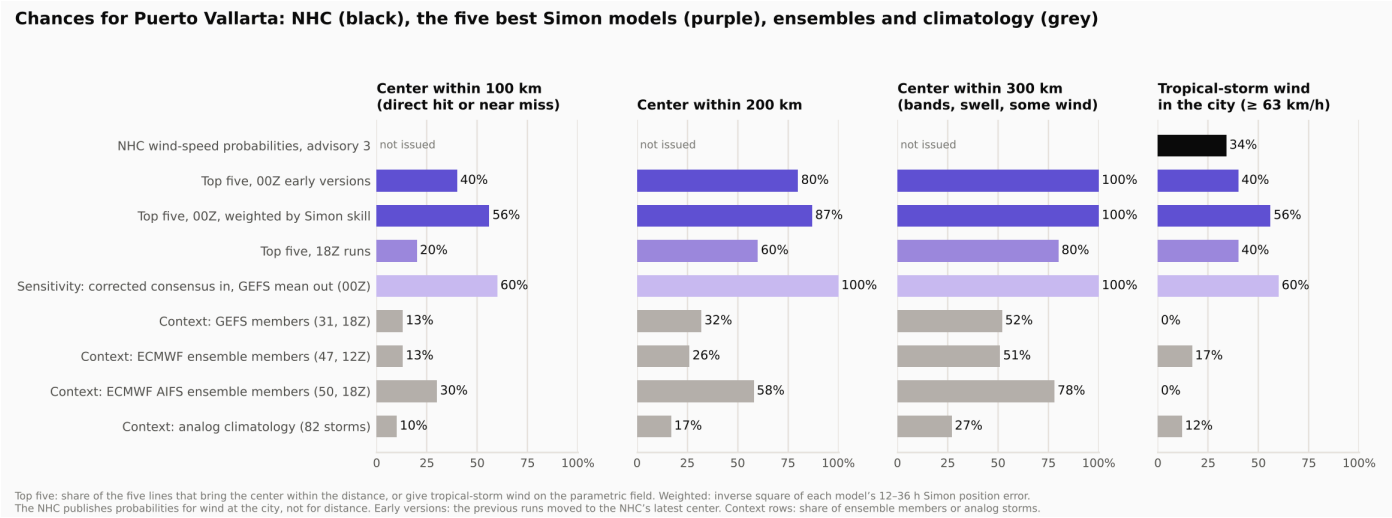


Figure 3. NHC probability of tropical-storm wind (black), the share of the five lines (purple) and of ensemble members and analog storms (grey) that bring the center within 100, 200 and 300 km or give tropical-storm wind in the city.

Outcome	NHC	Five, 00Z	Five, 18Z	AIFS ens. / ECMWF ens. / GEFS / climo	Our reading
Center within 300 km	–	100%	80%	78 / 51 / 52 / 27%	about 4 in 5
Center within 100 km	–	40%	20%	30 / 13 / 13 / 10%	about 1 in 3
Tropical-storm wind (≥ 63 km/h) in the city	34%	40%	40%	0 / 17 / 0 / 12%	about 1 in 3
Strong tropical-storm wind (≥ 93 km/h)	10%			–	about 1 in 10
Hurricane wind (≥ 119 km/h) in the city	2%			–	low

Climatology: 82 storms since 1949 passed within 500 km of Trudy's position between 12 September and 11 November; 22 came within 300 km of Vallarta, 8 within 100 km. The closest: Narda 2019 (11 km), Hernan 1996 (17 km), Maggie 1966 (24 km), Lorena 2019 (51 km), Virgil 1992 (68 km), Otis 1981 (70 km), Priscilla 1971 (82 km), Olaf 2003 (84 km), Patricia 2015 (118 km), Kenna 2002 (122 km).

4. The three addresses

Four columns. Least, most likely and worst are lines from the five best Simon models ("most likely" is their median by closest approach; "worst" the line with the strongest wind at the terrace). The NHC column turns the advisory's wind probabilities for Puerto Vallarta into a median and a 1-in-10 wind, times the same exposure factors (0.89 for the Amapas terrace with a land wind, 0.77 for the sheltered mid floor, 0.70 for the street-level house); its timing is from the official line. Times are Vallarta time.

	Least **GEFS ensemble mean 00Z: 292 km southwest at Sun 18 Oct 18:00, 37 kt**	**Most likely (five)** **Canadian ensemble mean 00Z: 166 km southwest at Sun 18 Oct 19:00, 43 kt**	**NHC advisory 3** **landfall near Manzanillo Saturday evening; center 38 km southwest at Sun 18 Oct 11:00**	**Worst (five)** **Google DeepMind cyclone model 00Z: 64 km southwest at Sun 18 Oct 08:00, 70 kt after a 110-kt peak**
Swell on the coast	–	Sat 17 Oct midnight → Sun 18 Oct midnight	Fri 16 Oct 06:00 → Mon 19 Oct midnight	Fri 16 Oct 06:00 → Mon 19 Oct 18:00
Rain bands	Sun 18 Oct 06:00 → Mon 19 Oct 18:00	Sun 18 Oct midnight → Tue 20 Oct 06:00	Sat 17 Oct noon → Mon 19 Oct midnight	Sat 17 Oct noon → Tue 20 Oct midnight
Tropical-storm wind in the city	–	–	34% chance, Sun 18 Oct midnight → Sun 18 Oct 19:00 if on the line	Sun 18 Oct midnight → Sun 18 Oct 15:00
Rain at the city (before terrain)	52 mm	102 mm	165 mm on the line	217 mm
D Terrace, Amapas, 8th floor	**17** km/h, gusts 24 Sun 18 Oct 16:00, wind from the E	**33** km/h, gusts 47 Sun 18 Oct 18:00, wind from the E	**48** km/h, gusts 67 1 in 10: 82 km/h, gusts 115	**74** km/h, gusts 103 Sun 18 Oct 07:00, wind from the E
Casa Francisca, Emiliano Zapata, 3rd floor	**15** km/h, gusts 21 Sun 18 Oct 16:00, wind from the E	**29** km/h, gusts 42 Sun 18 Oct 18:00, wind from the E	**41** km/h, gusts 60 1 in 10: 71 km/h, gusts 103	**64** km/h, gusts 92 Sun 18 Oct 07:00, wind from the E
El Pitillal house, street level	**13** km/h, gusts 19 Sun 18 Oct 16:00, wind from the E	**25** km/h, gusts 36 Sun 18 Oct 18:00, wind from the E	**37** km/h, gusts 54 1 in 10: 64 km/h, gusts 93	**56** km/h, gusts 81 Sun 18 Oct 07:00, wind from the E
What it means	Big surf and a wet Sunday; nothing more.	A hurricane passing offshore to the southwest on Sunday evening. Heavy rain Sunday, gusty wind, high surf on the open coast and the Malecón. Ordinary storm precautions.	A weakening hurricane crossing the Sierra and passing close to the bay on Sunday. Wind most likely below storm force in town but gusty; 1 in 10 for damaging gusts near 100 km/h on the exposed terrace; heavy rain on the slopes; river flooding likely in Pitillal; power cuts likely.	A hurricane 60–70 km southwest on Sunday morning. Tropical-storm wind from midnight to mid-afternoon Sunday, gusts near 100 km/h on the terrace, power and water cuts; 217 mm plus terrain puts the Pitillal and Cuale rivers in flood and risks slides on the Amapas and Conchas Chinas slopes. Pitillal: on higher ground by Saturday night.

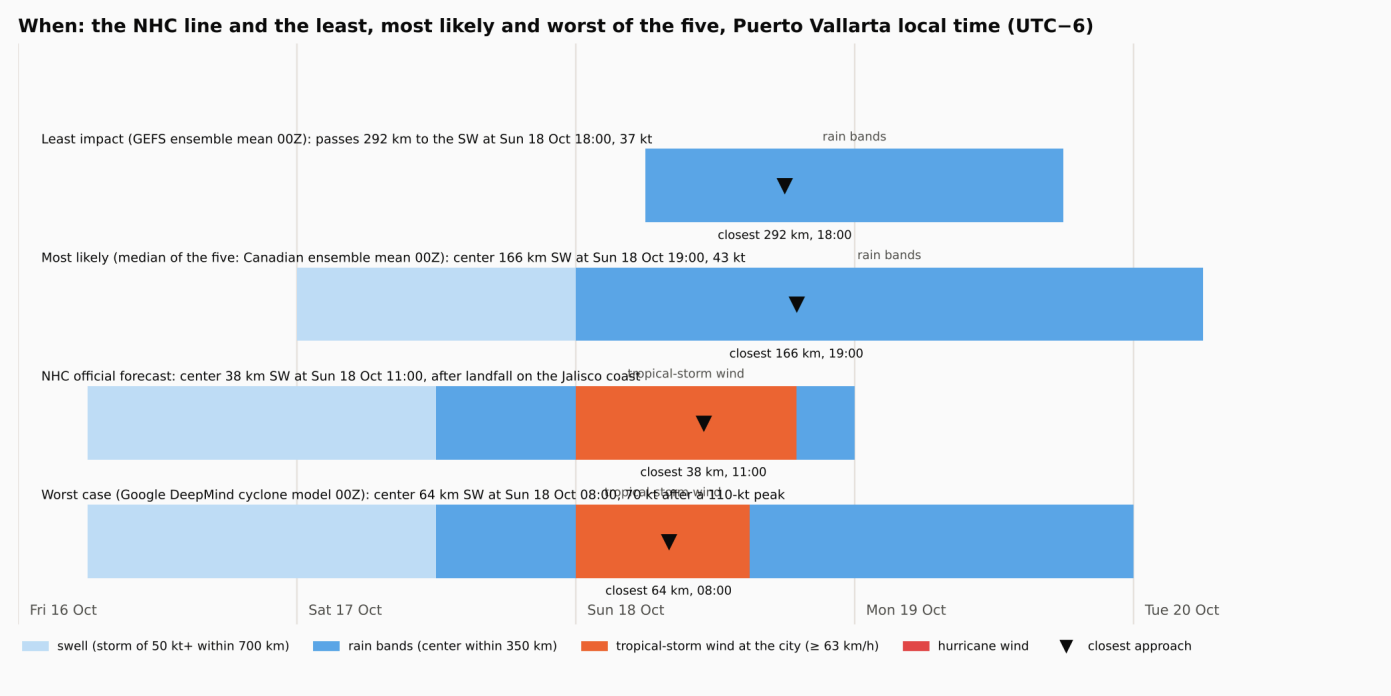


Figure 4. Timing of swell, rain bands and tropical-storm wind at the city on the NHC line and three of the five. The NHC row's wind bar assumes the center follows the line exactly; the NHC's own chance for that wind is 34%.

5. The GFS, out to 384 hours

The GFS run of 6 pm Monday (18Z) is the outlier for Trudy: it makes landfall in Michoacán on Saturday as a 90-kt hurricane and has it fall apart inland by Sunday (18.0N 103.8W, hour 123); the NHC calls this track unrealistic. The "turn" seen on GFS maps later in the run is a second storm, not Trudy: it forms south of Mexico near 10N 98W around Fri 23 Oct, 12 am, moves west-northwest, then turns north, peaks at 952 hPa (about 94 kt) 160 km southwest of Vallarta around Tue 27 Oct, 12 am, and crosses the coast just north of the bay that night. In that run, Puerto Vallarta gets about 312 mm of rain between 26 and 28 October and a peak wind of 40 km/h at the grid point. That is one run at 13–15 days, a storm that does not exist yet; at that range the GFS's skill for a specific storm is close to none. Treat it as a reminder that the season is not over, not as a forecast.

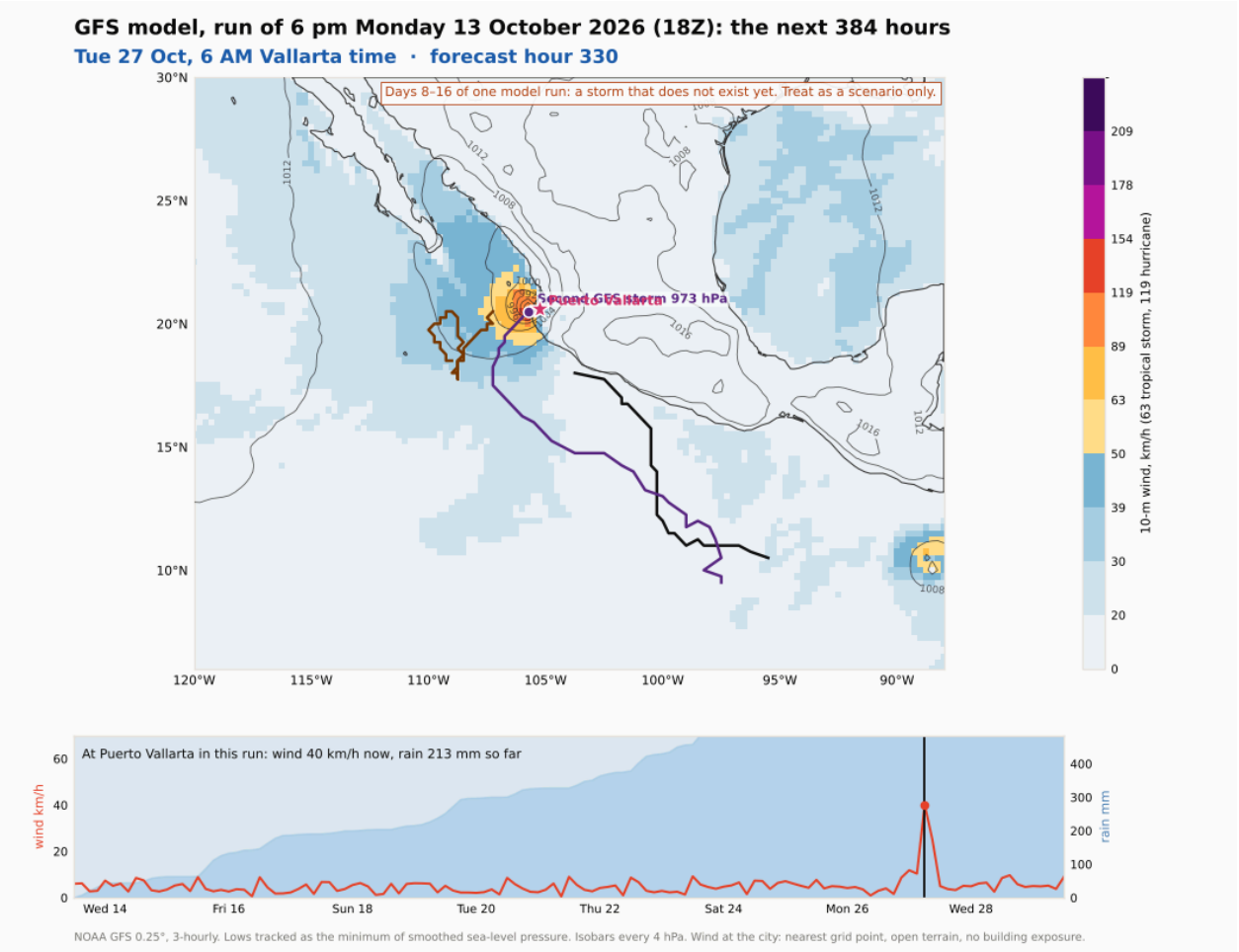


Figure 5. One frame of the animation (sent separately as a video): the GFS second storm off Cabo Corrientes on 26–27 October. Colours: 10-m wind; lines: sea-level pressure; bottom strip: the run's wind and rain at Puerto Vallarta.

6. When we will know

- **Wednesday:** NHC advisories at 3 am, 9 am, 3 pm and 9 pm Vallarta time; hurricane or tropical-storm watches for parts of the coast are likely. Watch the Puerto Vallarta line of the wind-speed probabilities.
- **Thursday:** about 72 hours out; NHC track errors drop to about 140 km. If the line still crosses the coast near Manzanillo and heads for the bay, prepare for the NHC column above; if it bends east inland or west offshore, the numbers fall.
- **Friday:** warnings 36 hours before tropical-storm winds. Simon's lesson: even inside 24 hours the line can shift 100 km; read the probabilities, not the line.

Simon, meanwhile (advisory 27): a tiny 95-kt hurricane 200 km west of Cabo Corrientes, re-intensifying and drifting west at 4 km/h, forecast to move near Las Islas Marías on Thursday and weaken; a Hurricane Watch is in effect for Las Islas Marías; the NHC's confidence is low. Its swell continues on the coast.

Appendix. Every aid

A1. 00Z cycle (official forecast and early versions)

Aid	Closest, km	Side	When (local)	kt then	Peak kt	Rain mm	Landfall
NHC official forecast	38	SW	Sun 18 11:00	87	115	165	Jalisco south coast, Sat 17 20:00, 112 kt
Google DeepMind (early)	64	SW	Sun 18 08:00	70	110	217	Jalisco south coast, Sat 17 21:00, 80 kt
ECMWF AIFS (AI, early) †	47	WSW	Sun 18 11:00	79	89	145	Jalisco south coast, Sun 18 09:00, 79 kt
Track consensus †	187	SW	Mon 19 02:00	76	89	107	–
Canadian ens. mean (early, 12 h)	166	SSW	Sun 18 19:00	43	52	102	–
GEFS mean (early)	292	SW	Sun 18 18:00	37	49	52	–
NHC corrected consensus	21	SW	Sun 18 11:00	79	123	347	Michoacán / Colima, Sat 17 17:00, 102 kt
GFS (early)	416	SE	Sun 18 18:00	21	87	8	–
UKMET (early, 12 h)	199	SW	Sun 18 18:00	48	49	87	–
NAVGEM (early)	196	SSW	Sat 17 18:00	61	89	80	Sinaloa, Tue 20 11:00, 54 kt
Canadian GEM (early, 12 h)	289	SSW	Sun 18 17:00	44	48	43	–
HAFS-A (early)	336	S	Mon 19 06:00	113	113	8	–
HAFS-B (early)	325	SSE	Mon 19 06:00	93	105	10	–
HWRF (early)	230	SSW	Mon 19 03:00	32	78	45	–
HMON (early)	106	WSW	Sun 18 10:00	78	85	98	–
COAMPS-TC (early)	224	WSW	Sun 18 16:00	126	129	32	–
Climatology-persistence	112	SSW	Sun 18 11:00	–	–	0	Michoacán / Colima, Sat 17 03:00, 0 kt

A2. 18Z runs

Aid	Closest, km	Side	When (local)	kt then	Peak kt	Rain mm	Landfall
Google DeepMind	103	WSW	Sun 18 08:00	72	117	169	–
ECMWF AIFS (AI) †	93	WSW	Sun 18 noon	77	92	110	Sinaloa, Sun 18 22:00, 70 kt
Track consensus †	255	SSW	Mon 19 06:00	67	92	60	–
Canadian ens. mean (early)	199	SW	Sun 18 20:00	44	54	82	–
GEFS mean	314	SW	Sun 18 18:00	37	51	39	–
NHC corrected consensus	57	SSW	Mon 19 02:00	85	119	325	Michoacán / Colima, Sun 18 05:00, 100 kt
GFS	392	SE	Mon 19 06:00	14	98	6	Michoacán / Colima, Sun 18 23:00, 15 kt
UKMET (early)	240	SW	Sun 18 19:00	48	50	61	–
NAVGEM	226	SW	Sat 17 20:00	51	93	65	–
HAFS-A	444	SSE	Sun 18 18:00	112	118	3	–
HAFS-B	443	SSE	Sun 18 18:00	95	110	3	–
HWRF	318	S	Sun 18 18:00	45	86	16	–
HMON	140	WSW	Sun 18 noon	75	93	60	–
COAMPS-TC	243	WSW	Sun 18 16:00	124	134	16	–
Climatology-persistence	203	S	Sun 18 noon	–	–	0	–

† intensity from the NHC intensity consensus. ECMWF IFS 12Z: 323 km SW at Tue 20 Oct 06:00. Ensembles: GEFS 18Z (31 members), ECMWF 12Z (47 members tracked as disturbance 71E), AIFS 18Z (50 members). Method as in the earlier Trudy brief: hourly interpolation, Natural Earth land, parametric wind with model radii where available, R-CLIPER rain, Simon exposure factors. Climatology: HURDAT2 1949–2025 centred on 11.0N 96.7W.

Code and data: hurricane-simon/trudy in branch claude/hourly-forecast-animation-m4v1bj of the vallarta-weather repository. Not an official forecast. Follow the SMN, the NHC and Protección Civil. Emergencies: 911.